Ammonia is a high-density H_2 energy carrier.

A one-page evidence brief for the NH_3 node in the Global Systems Tensor Network. Figures cited to primary literature.

12.7 MJ / L

liquid NH_3 · −33 °C · 10 bar

vs **8.49** liquid H_2 **+50%** per litre

MORE ENERGY PER LITRE

12.7 MJ/L

vs 8.49 liquid H_2

Liquid NH_3 stores 50% more chemical energy per litre than liquid H_2. 1.7× more H_2 per m³ (121 vs 71 kg-H_2/m³).

ACS Publications, 2023

EASIER TO LIQUEFY & SHIP

−33 °C

vs −253 °C for LH_2

NH_3 liquefies 220 °C warmer than H_2, cutting liquefaction energy from ~10–12 to ~0.5 kWh per kg of H_2.

Patsnap 2026 · ARIEMA 2025

INFRASTRUCTURE EXISTS

180 Mt/yr

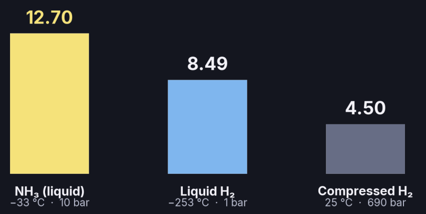
Haber–Bosch today

~200 carriers and 150+ ports already handle NH_3. Synthesizable from water, air, and electricity — around fossil chokepoints.

Heise / IFA 2024 · ARIEMA 2025

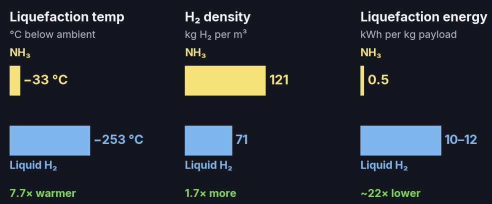
VOLUMETRIC ENERGY DENSITY (MJ/L)

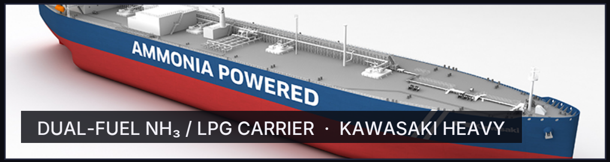
ACS Publications · 2023



STORAGE & HANDLING

Patsnap 2026 · ARIEMA 2025





DUAL-FUEL NH_3 / LPG CARRIER · KAWASAKI HEAVY

GREEN NH_3 ELECTROLYZER PLANT · NEL HYDROGEN

WHY NH_3 BELONGS ON THE ENERGY-DENSITY NODE

Energy density is the #1 systemic lever in the network (12 connections, more than any other node). Ammonia raises it without trading climate, agriculture, or supply-chain resilience. At 12.7 MJ/L liquid (−33 °C, ~10 bar), NH_3 carries 50% more energy per litre than liquid H_2. Synthesizable from water + air + electricity — the structural fix the model points to.

SOURCES

[1] ACS Publications, Advances and Perspectives of H_2 Production from NH_3, Energy & Fuels (2023). NH_3 12.7 MJ/L; LH_2 8.49 MJ/L; NH_3 storage 9.9 bar / 25 °C. pubs.acs.org

[2] Patsnap, Ammonia as hydrogen carrier: 2026 technology landscape (2026). 121 vs 71 kg-H_2/m³; liquefaction 0.5 vs 10–12 kWh/kg. patsnap.com

[3] Heise / IFA: ~180 Mt/yr NH_3 via Haber–Bosch (2024). heise.de

[4] ARIEMA (2025): 150+ ports handle NH_3 today. ariema.com